

Welcome to the BIAI project,

As per the schedule, we are starting the project the day after tomorrow (Wednesday), which will run until the end of June, so please make sure you attend.

1. What will the project involve?

The aim of your project will be to implement a genetic algorithm to solve a problem of your choice. In other words, you will need to get the programme to learn how to solve the problem (e.g. find the shortest path) on its own. You can see how such a genetic algorithm works in practice in this

video: <https://www.youtube.com/watch?v=mA8z0GndiYI>

2. How to carry out the project?

The project will be carried out in pairs. The project will consist of two tasks:

- At the start, each group will choose one task as per your own selection or choice. The aim of the first task is to familiarise you with how genetic algorithms work.
- The second task will involve adapting a genetic algorithm to one of the OpenAI test environments available in the library: <https://gymnasium.farama.org/> The aim of the second task will be to use a genetic algorithm in an environment with a graphical interface to solve a practical problem.
- Finally (during the last class), there will be presentations. Each group will prepare a presentation demonstrating how the genetic algorithm works on a problem of their choice.

3. Tools used in the project

The project will be developed in your choice programming platform (e.g., Python etc.). As an example, the <https://pygad.readthedocs.io/en/latest/>, library will be used to implement the genetic algorithm; this library provides a simple interface for defining the components of a genetic algorithm, such as population, selection, crossover and mutation. All plots will be created using the Matplotlib library. Ultimately, I would like your projects to be hosted on GitHub.